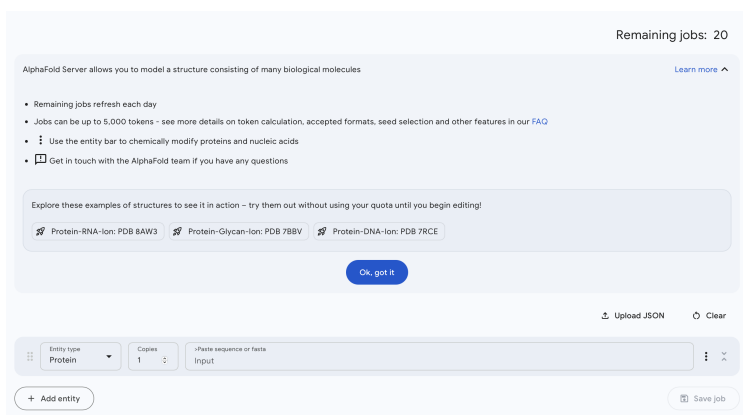


CHEM 515 Practice Session 6.2: Protein structure prediction using the AlphaFold server

Name: _____

In this additional part of **Practice 6**, we are going to use the AlphaFold server to predict the structure of fasciculin (PDB ID: 1FSC), a small protein found in the venom of green mamba snake. Please follow the instructions as follows:

1. Open the RCSB PDB website (<https://www.rcsb.org>). Search for PDB ID “1FSC”.
2. Using the dropdown menu of “Download Files”, download two files to your computer: (i) the “**FASTA sequence**”, which contains the sequence information; (ii) the “**PDB Format**”, which is the PDB file for this protein.
3. Open the AlphaFold server (<https://alphafoldserver.com>). For the first-time log-in, you will need to sign up (the server is free for academic use; please specify your organization as “San Diego State University”) and agree upon the Terms and Conditions. After you log in, you should see a page as shown below:



Note that a single user can submit at most 20 jobs to the server per day.

4. The fasciculin is a simple protein containing only one chain (no ligand). From the downloaded FASTA file, you can find the sequence of this chain on the 2nd line. Use that as the input for the job on AlphaFold server. The Entry type should remain to be “Protein”.
5. The job is now ready for submission. Rename it as “1FSC” on the job preview page. It should be able to finish in 1–2 minutes on the server.
6. Once the job finishes, you can click on it to see the predicted 3D structure and an estimation of potential errors of the prediction. Click “Download” on this page to download the results as a ZIP file. Unzip the downloaded file on your computer, under which you will find a file “fold_1FSC_model_0.cif”. This corresponds to the predicted structure that is of the highest score.
7. Use VMD to open fold_1FSC_model_0.cif and the original PDB file (“1fsc.pdb”) as two “New Molecules”. Set “1fsc.pdb” as the molecule at the top molecule. This can be achieved by double-clicking under the capital letter “T” in the “1fsc.pdb” row under the “VMD Main” window.

8. Under “**Graphics** → **Representations**”, create two representations for the two structures based on the specifications below:

Selected Molecule	Drawing Method	Coloring Method	Material
1fsc.pdb	NewCartoon	Secondary Structure	Transparent
fold_1FSC_model_0.cif	NewRibbons	Secondary Structure	Opaque

Note that for both molecules, the field “**Selected Atoms**” should simply be “**All**”.

9. We are now going to overlay the two protein structures and calculate the RMSD. This can be achieved using “**Extensions** → **Analysis** → **RMSD Calculator**”, Open that up, and type “**protein**” in the box at top left of the “**RMSD Tool**” window. Then click the “**Align**” button. The VMD display should now present the AlphaFold-predicted structure superimposed on the downloaded PDB structure. Turn off the Axes, and change the view to “**Orthographic**”. Then save an image for the overlaid protein structures.
10. Finally, under the “**RMSD Tool**” window, click on the “**RMSD**” button. This will calculate the RMSD between the two structures. The result will be displayed on that window.

Files to submit:

1. The image of overlaid protein structures generated in **Step 9** above (**0.8 pt**)
2. A screenshot of the report of the completed job generated by the AlphaFold server, including the predicted 3D structure and the heatmap of estimated errors (**0.3 pt**)
3. A screenshot of the “**RMSD Tool**” window of VMD that displays the value of the calculated RMSD. (**0.4 pt**)